As in, “research papers spanning dozens of pages” complicated. Describing them in details goes beyond the scope of this chapter, but we’ll briefly describe a few prominent ones, and provide pointers for the interested readers[17].

Be aware that the effort to implement these algorithms might be consistent; also keep in mind that if your constraints can be relaxed, and you can accept an algorithm that provides a reasonable embedding, even without guarantees that it will be planar, you can use simpler heuristics to generate embeddings (more on this later).

As we mentioned, the first linear-time algorithm for planarity testing was developed in 1974 by Hopcroft and Tarjan[18], improving a previously-developed[19] $O(|V|^2)$ variant; their idea is based on vertex addition, so the algorithm starts bottom-up, keeping the possible planar embeddings for each incrementally-built *induced subgraph*[20] of the original one.

As we had previously mentioned, this strategy defines a different approach to recursion: bottom-up rather than top-down, incremental and not divide-and-conquer, but above all, by carefully re-constructing the original graph one vertex at the time, it avoids analyzing all sub-graphs, and only performs a linear number of steps.

The key is that, while adding vertices, the algorithm keeps track of the possible embeddings of the sub-graph.

Fast-forward to 2004, when a brand-new approach[21] was developer by Boyer and Myrvold: it’s an edge-addition method, so incrementally adding edges instead of vertices,